

Some Popular Models

Specifications	Tektronix TBS1000C Series	Agilent DSO1000B	Teledyne Lecroy WaveAce 1000	GW Instek GDS2000A Series	Aplab D36000
					
Analogue bandwidth	25 MHz to 150 MHz	50 MHz to 150 MHz	40 MHz to 100 MHz	200 MHz	200 MHz
Rise time			8.8 ns to 3.5 ns	1.75 ns	1.75 ns
Sample rate	500 MSa/s to 1.0 GSa/s	1 GSa/s	1 GSa/s	2 GSa/s	1 GSa/s
Record length	2.5k points	16k points	1M points	2M points	4k points
External trigger	Yes		Yes	Yes, $\pm 15V$	Yes, $\pm 1.6V$
Trigger types	Edge, pulse width, video	Edge, pulse width, video, external, composite view and alternate	Edge, pulse width, video, slope, alternate	Edge, pulse width, video, pulse runt, rise and fall, alternate	Edge, video, pulse
Maximum waveform capture rate		200 Wfm/s		80,000 Wfm/sec	2000 Wfm/s
Number of channels	2	2	2	4	2
Probes specification	100MHz passive probe	150MHz/300MHz passive probes	10:1, 1:1 switchable passive probe	250MHz (10:1/1:1) switchable passive probe	10:1, 1:1 switchable passive probe
Vertical resolution	8 bits	8 bits	8 bits	8 bits	8 bits
Vertical sensitivity	2 mV/div to 5V/div	2 mV/div to 10V/div	2 mV/div to 10V/div		2 mV/div - 5V/div
Vertical accuracy	$\pm 3\%$	5 mV/div		± 3 per cent	± 3 per cent
Vertical zoom	Vertical expand	Vertical expand		None	None
Time base range	5 ns to 50s/div	2 ns/div to 50s/div	2.5 ns/div to 50s/div (for 100MHz model)	1 ns/div to 100 s/div (1-2-5 increments); Roll: 100 ns/div - 100s/div	1 ns/div- 50s/div
Time base accuracy	50 ppm	± 50 ppm over 1 ms		20 ppm over any ≥ 1 ms time interval	50 ppm
Horizontal zoom	Horizontally expand	Yes		None	Horizontal zoom x10
Automated waveform measurement	Yes	Yes	Yes	Yes	Yes
Waveform math	Yes, including FFT	Yes, including FFT	Yes, including FFT	Yes, including FFT	Yes, including FFT
I/O Interfaces	USB, GBIP (optional)	USB (host and device)	USB (host and device)	RS232C, USB port, Ethernet port, Go-NoGo BNC, SVGA video port, GPIB	USB, RS232, USB host for flash drive
Display/Size	145mm active colour TFT	145mm colour LCD display with wide angle viewing	178mm colour display	203mm, 800×600 high-resolution TFT LCD display	145mm TFT LCD, 320×234 pixels
Overall dimensions	32.6×15.8×12.4 cm ³ ; 2 kg	30.3×15.4×13.3 cm ³ ; 3.03 kg (A model)	16.3×31.3×11.6 cm ³ ; 2.78 kg	38×22×14.5 cm ³ ; approx. 4.2 kg	30×14.5×31.2 cm ³ ; 4.5 kg
Safety	300 Vrms CATII	CAT 1300 Vrms, 400 Vpk; transient overvoltage 1.6kVpk	400Vpk, CATI	300V (DC+AC Peak), CAT I	CAT II

ments, you need a rugged scope. Below-mentioned specifications can help you judge an oscilloscope for your work environment.

Display. Smaller displays are very uncomfortable. It becomes very difficult to analyse some high-frequency signals or bus on a small display. Look for an oscilloscope that has com-

paratively bigger display size. Colour displays are pleasing to the eyes and make it easy to distinguish signals by representing them in different colours. Entry-level oscilloscopes are generally available with a 14.4cm (5.7-inch) display.

Dimensions and weight. Dimensions and weight should be carefully

selected depending on your work environment. If you work in the field, you will not be able to carry around a big oscilloscope. Taking measurements with it will also be very uncomfortable. So look for a small-size and light-weight oscilloscope. ●

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